John Abascal

COMPUTER SCIENCE PHD STUDENT | AI PRIVACY, SECURITY, AND SAFETY

abascal.j@northeastern.edu | johnabascal.com | johnmath | johnabascal

Education

Doctor of Philosophy in Computer Science
Boston, MA

NORTHEASTERN UNIVERSITY — *Sep. 2021 - Present (Expected Jul. 2026)*

- **Research Topics:** Adversarial Machine Learning, Privacy Preserving Machine Learning, Differential Privacy, AI Safety
- **Advisors:** **Alina Oprea** and **Jonathan Ullman**
- **GPA:** 3.94/4.00

Bachelor of Science in Mathematics
Tallahassee, FL

FLORIDA STATE UNIVERSITY — *Aug. 2017 - May 2021*

- **Honors Thesis:** ”A Standardized Approach to Studying the Double Descent Phenomenon”
- **Advisors:** **Alec Kercheval** and **Nathan Crock**
- **Honors:** Honors in the Major, Florida Academic Scholar, Hispanic Scholarship Fund

Skills

Languages	Python, C++, Swift, C, JavaScript, MATLAB, LaTeX; Spanish
Tools & Technologies	PyTorch, Opacus, Hugging Face, scikit-learn, NumPy, Asyncio, Pandas, OpenCV, Matplotlib, PIL, UIKit, Linux, GCP

Experience

Google Research
Cambridge, MA

STUDENT RESEARCHER (MACHINE UNLEARNING) — *Nov. 2024 - Aug. 2025*

- Led a research effort investigating the practicality of certified unlearning and authored an academic paper to communicate our contributions.
- Wrote a test bench in PyTorch and Hugging Face to measure trade-offs between cost, accuracy, and privacy of several certified unlearning algorithms for vision models and large language models.

LinkedIn
Sunnyvale, CA; New York, NY

APPLIED RESEARCH INTERN (DATA PRIVACY) — *May 2023 - Aug 2024*

- (Summer 2023) Developed inference attacks with theoretical guarantees to measure privacy leakage across LinkedIn’s analytics.
- (Summer 2024) Measured leakage in pre-production large language models by mounting membership-inference and data extraction attacks.
- Communicated risks and differential privacy-based mitigations to foster the adoption of privacy-enhancing technologies.

Intuit
Mountain View, CA

SOFTWARE ENGINEERING INTERN (IOS) — *May 2020 - Aug 2021*

- (Summer 2020) Created a modular service layer using the Apollo Client that can be used by future QuickBooks applications.
- (Summer 2021) Developed new features for the Money by QuickBooks app and modular UI assets for all of Intuit’s iOS applications.

Publications

Black-Box Privacy Attacks on Shared Representations in Multitask Learning **(ICLR 2026)**
John Abascal, Nicolás Berrios, Alina Oprea, Jonathan Ullman, Adam Smith, Matthew Jagielski
2025

Should We Forget About Certified Unlearning? Evaluating the Pitfalls of Noisy Methods **(Preprint)**
John Abascal, Eleni Triantafillou, Matthew Jagielski, Nicole Mitchell, Peter Kairouz
2025

Phantom: General Trigger Attacks on Retrieval Augmented Language Generation **(ACM TAISAP)**
Harsh Chaudhari, Giorgio Severi, **John Abascal**, Matthew Jagielski, Christopher A. Choquette-Choo, Milad Nasr et al.
2024

TMI! Finetuned Models Leak Private Information from their Pretraining Data **(PETS 2024)**
John Abascal, Stanley Wu, Alina Oprea, Jonathan Ullman
2023

SNAP: Efficient Extraction of Private Properties with Poisoning **(IEEE S&P 2023)**
Harsh Chaudhari, **John Abascal**, Alina Oprea, Matthew Jagielski, Florian Tramèr, Jonathan Ullman
2022

Current Research Projects

Measuring Privacy Leakage Through Distillation
Boston, MA

KHOURY COLLEGE OF COMPUTER SCIENCES AT NORTHEASTERN UNIVERSITY — *Oct. 2025 - Present*

- Designing training data extraction attacks to measure leakage of implicitly learned information from distilled LLMs.
- Exploring the use of extraction attacks to audit whether a model was distilled from a proprietary LLM.